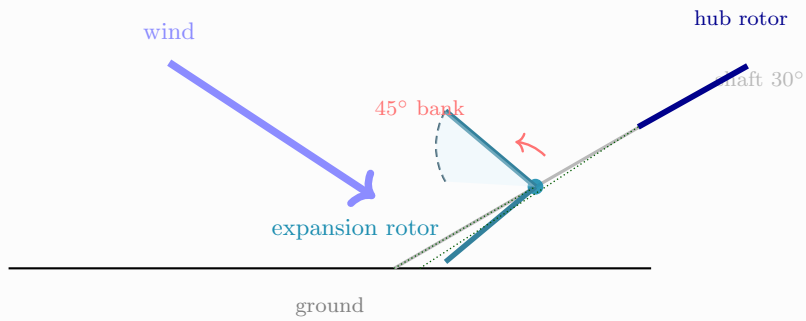
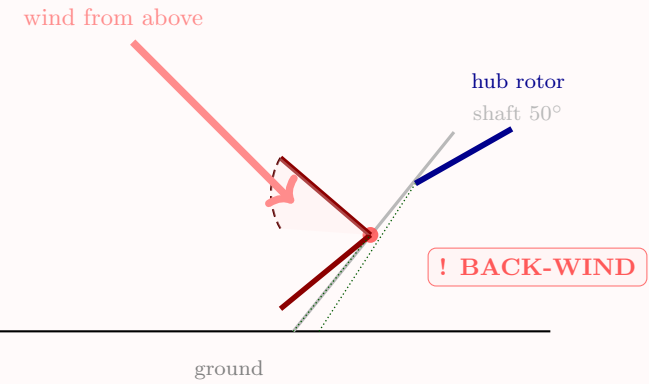


Expansion Rotor — Normal Operation vs. Pitch Depower

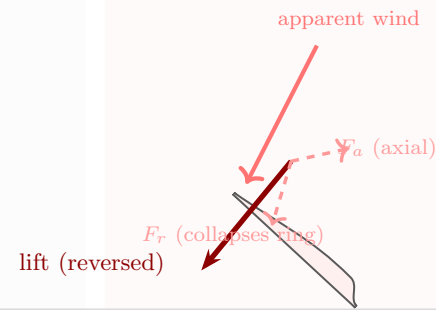
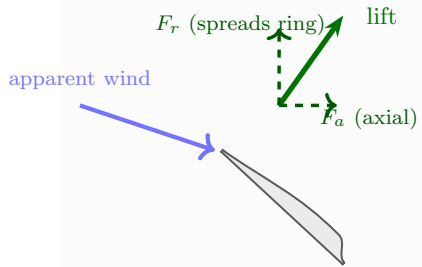
Normal: 30° elevation, wind from front



Depower: 50° elevation, wind hits from above



pitch depower



At 45° bank, pitch depower reverses radial force — tethers pull inward, r_{eff} drops, shaft can buckle. Dynamic simulation required.